# Sparse Gaussian Gradient Code

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***Abstract*—Gradient coding is a distributed computing technique aiming to provide robustness against slow or non-responsive computing nodes, known as stragglers, while balancing the computational load for responsive computing nodes. Among existing gradient codes, a construction based on combinatorial designs, called BIBD gradient code, achieves the best trade-off between robustness and computational load in the worst-case adversarial straggler setting. However, the range of system parameters for which BIBD gradient codes exist is limited. In this paper, we overcome this limitation and propose a new probabilistic gradient code, termed Sparse Gaussian (SG) gradient code. The encoding matrix of the proposed SG gradient code is generated from a carefully chosen correlated multivariate Gaussian distribution, masked by Bernoulli random variables to reduce computational load. With high probability, the proposed gradient code achieves a similar worst-case error performance compared to the BIBD gradient code (when such a code of the same parameters exists) and outperforms several other existing gradient codes, including Fractional Repetition gradient codes and Bernoulli gradient codes. Moreover, it further extends the range of system parameters over existing BIBD and soft BIBD gradient codes, making it a promising solution for distributed computing tasks.**

## I. Introduction

Due to the ever-increasing computing demands in various applications such as machine learning and cloud computing, it becomes essential for large systems to perform their computations in a distributed manner. In theory, distributed algorithms can substantially speed up computations compared to centralized algorithms. However, these speed-up gains may not be realized in practice due to the presence of underperforming or non-responsive computing nodes, termed as stragglers [1]. Thus, how to distribute computations that are robust against stragglers is a major challenge in the design of distributed computation systems. In pursuit of this objective, numerous strategies have been developed to mitigate the impact of stragglers. These include duplicating tasks across computing nodes [2] and, where feasible, excluding stragglers from the computation process [3]. The primary challenge in these approaches lies in how to balance communication efficiency against straggler tolerance [4].

To address this challenge, methodologies grounded in coding theory have been explored. *Gradient Coding* (GC) has emerged as an important tool. It involves dividing each training data into partitions and distributing them to certain computing nodes known as workers. These workers compute and send a linear combination of gradients to the master node, aggregating these to recover the gradient sum accurately. The utmost goal of GC is to recover the gradient sum exactly. Several gradient codes have been proposed under the exact recovery criterion, such as *Fractional Repetition* gradient codes. However, a significant drawback of exact recovery is its potentially high computing workload as it increases with the number of stragglers.

Though exact recovery is desired, approximate recovery may be sufficient in many practical applications, such as stochastic gradient descent algorithms [5]. Instead of exactly recovering the gradient sum, approximate recovery finds an approximation of the gradient sum [6]. The $\ell_2$-norm of the difference between the exact gradient sum and the approximate gradient sum (i.e. squared error) is a common performance metric. There are two common assumptions for the stragglers: (1) the stochastic stragglers, where computing nodes are straggled independently with some probability, and (2) adversarial stragglers, where stragglers can deliberately disrupt any set of $S$ computing nodes. Throughout this paper, we assume the adversarial straggler setting, which is more realistic for practical systems where obtaining a statistical estimation of the stragglers in real time is difficult. Under this assumption, we maximize the squared error over all possible straggling patterns and refer to it as *the worst-case squared error*.

In the literature, many approximate gradient codes with small worst-case squared errors are introduced, for example, [2], [7], [6], [8], [9]. Among these, one research direction in approximate gradient codes involves the utilization of the bi-adjacency matrix from expander graphs to develop gradient codes [7]. Expander graphs are favourable for gradient coding due to their strong connectivity, which enhances the robustness against stragglers. It is shown that the upper bound of the squared error is proportional to the second largest eigenvalue of the normalized adjacency matrix for the expander graph [7]. This makes Ramanujan graphs a good candidate as they provide small second largest eigenvalues. As a special type of Ramanujan graphs, *Balanced Incomplete Block Designs* (BIBDs) achieves the best error performance with a closed-form expression for the worst-case error. Moreover, the error depends on the straggling pattern only through the number of stragglers, not the specifics of the straggling pattern [8].

Unfortunately, most existing constructions for expander graphs require extensive computer search [10] and the set of system parameters for which expander graphs exist is limited [11]. In particular, the allowable parameters of BIBDs are notably limited [12]. Consequently, alternative gradient codes have been developed to emulate BIBD gradient codes by generating probabilistic gradient codes, rather than deter-

ministic ones. This is termed soft BIBD gradient codes in [9]. It is important to note that while these designs expand the range of parameters of BIBDs, they are still restricted to binary gradient codes, that is, the encoding matrix has binary entries. Moreover, the construction of the encoding matrix involves sampling from a complicated high-dimensional distribution, which may be difficult to implement.

This motivates us to ask the following question: *by allowing a real-valued encoding matrix instead of a binary one, is it possible to simplify the gradient code construction, broaden the range of system parameters, and retain a similar error performance as BIBD gradient codes?* In this study, we answer the question affirmatively by proposing a probabilistic gradient code termed *Sparse Gaussian* (SG) gradient code. Its construction contains two steps: We first choose a correlated multivariate Gaussian distribution and generate all rows of the encoding matrix i.i.d. according to this distribution. Then, we multiply each entry in the encoding matrix by an independent Bernoulli random variable to control the computation load. Unlike sampling from a complicated high-dimensional distribution as in the construction of soft BIBD gradient codes, the proposed SG gradient code only requires sampling from a multivariate Gaussian distribution. This makes it much easier to obtain in practice. Furthermore, this method allows a more extensive range of system parameters compared to BIBD gradient codes and their variant, soft BIBD gradient codes, as illustrated in Fig 1 in Section IV. Lastly, the proposed joint distribution of the encoding matrix mimics the properties of the BIBD encoding matrix, leading to an error performance comparable to that of symmetric or dual BIBD gradient codes with high probability, as shown in Theorem 2 and Fig. 2 in Section V.

## II. Problem Formulation

Let $\boldsymbol{E}$ be a matrix, $\boldsymbol{E}^T$ be its transpose and $\boldsymbol{E}_{ij}$ be its $(i,j)$-th entry. For positive integers $n$ and $k$, we denote an $n \times n$ identity matrix by $\boldsymbol{I}_n$, an $n \times k$ all-one matrix by $\boldsymbol{J}_{n\times k}$ and an $n \times k$ all-zero matrix by $\boldsymbol{0}_{n\times k}$. Let $\boldsymbol{v} \in \mathbb{R}^n$ be a column vector and $\boldsymbol{1}_n$ be an all-one column vector. The $\ell_p$-norm is represented by $\|\cdot\|_p$.

Consider a computing machine that includes a master node, $N$ workers, and a training set $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^M$, where $N, M \in \mathbb{N}^+$, $\boldsymbol{x}_i \in \mathbb{R}^d$, and $y_i \in \mathbb{R}$. The master node intends to train a model using gradient descent by distributing the gradient updates amongst the workers. Specifically, we wish to find the $\boldsymbol{w} \in \mathbb{R}^d$ that minimizes the empirical risk, i.e., $\min_{\boldsymbol{w}\in\mathbb{R}^d} \frac{1}{M}\sum_{i=1}^M \ell(\boldsymbol{x}_i, y_i; \boldsymbol{w})$, where loss function $\ell(\boldsymbol{x}_i, y_i; \boldsymbol{w})$ measures the accuracy of prediction made by $\boldsymbol{w}$ on $(\boldsymbol{x}_i, y_i)$.

The optimization problem can be addressed using the gradient descent algorithm. Specifically, the gradient of the loss at the current model $\boldsymbol{w}^{(t)}$ is computed as $\sum_{i=1}^M \nabla\ell(\boldsymbol{x}_i, y_i; \boldsymbol{w}^{(t)})$, and the model is updated iteratively based on this gradient. The gradient sum computation will be a bottleneck when dealing with large training data size $M$. To address this challenge, we distribute gradient computations across multiple workers, where each worker computes some subsets of the $M$ gradients whose sum is returned to the master to update the model.

In a distributed gradient computation setting, the training set $\mathcal{D}$ can be partitioned to $K$ disjoint partial data set $\mathcal{D}_i$ for $i \in [K]$, where $|\mathcal{D}_i| = M/K$. Define a partial gradient vector corresponding to $\mathcal{D}_i$ as $\boldsymbol{d}_i \triangleq \sum_{(\boldsymbol{x}_i, y_i)\in\mathcal{D}_i} \nabla\ell(\boldsymbol{x}_i, y_i; \boldsymbol{w})$.

In the context of $S$ stragglers, a gradient coding scheme contains two steps: distributing data to workers based on an encoding matrix and approximating the gradient sum from the received $N - S$ partial gradients.

A gradient code can be characterized by an encoding matrix $\boldsymbol{E} \in \mathbb{R}^{K\times N}$, where row $i \in [K]$ corresponds to data partition $\mathcal{D}_i$ and column $j \in [N]$ corresponds to worker $j$. Worker $j$ will return $\sum_{i=1}^K \boldsymbol{E}_{ij}\boldsymbol{d}_i$ where $\boldsymbol{E}_{ij}$ is the linear combination coefficient of the partial gradient corresponding to $(\boldsymbol{x}_i, y_i)$.

When each data piece is distributed to an equal number of $R$ workers (the number of non-zero entries in each row is $R$) and each worker computes an equal number of $L$ partial gradients (the number of non-zero entries in each column is $L$), we denote the gradient code as an $(N, K, L, R)$-GC.

The goal of the master node is to recover the gradient sum

$$\sum_{i=1}^K \boldsymbol{d}_i = \boldsymbol{D}\boldsymbol{1}_K,$$

where $\boldsymbol{D} = [\boldsymbol{d}_1, \boldsymbol{d}_2, \ldots, \boldsymbol{d}_K]$.

Let $\mathcal{F} \subset [N]$ be the set of non-straggling workers. We have $|\mathcal{F}| = N - S$. If $\mathcal{F} \neq \emptyset$, $\boldsymbol{E}$ is reduced to a sub-matrix $\boldsymbol{E}_{\mathcal{F}} \in \mathbb{R}^{K\times(N-S)}$ with columns indexed by $\mathcal{F}$. Therefore, the master node will receive $\boldsymbol{D}\boldsymbol{E}_{\mathcal{F}}$ where each entry is a partial gradient sum by a non-straggling worker. To approximate the gradient sum, the master node finds a linear combination $\boldsymbol{v} \in \mathbb{R}^{N-S}$ of the received gradient vector $\boldsymbol{D}\boldsymbol{E}_{\mathcal{F}}\boldsymbol{v}$, which is also referred as a decoding vector of $\boldsymbol{E}_{\mathcal{F}}$.

Thus, the difference between the target sum and the approximate sum can be expressed as

$$\boldsymbol{D}\boldsymbol{E}_{\mathcal{F}}\boldsymbol{v} - \boldsymbol{D}\boldsymbol{1}_K = \boldsymbol{D}(\boldsymbol{E}_{\mathcal{F}}\boldsymbol{v} - \boldsymbol{1}_K).$$

Since $\boldsymbol{D}$ is unknown to us, the goal is to find a decoding vector that minimizes the squared $\ell_2$-norm of the difference. So, given an encoding matrix $\boldsymbol{E} \in \mathbb{R}^{K\times N}$ and non-straggler pattern $\mathcal{F} \subset [N]$, the optimal decoding vector is

$$\boldsymbol{v}_{\text{opt}}(\boldsymbol{E}_{\mathcal{F}}) \triangleq \arg\min_{\boldsymbol{v}\in\mathbb{R}^{|\mathcal{F}|}} \|\boldsymbol{E}_{\mathcal{F}}\boldsymbol{v} - \boldsymbol{1}_K\|_2^2.$$

The performance metric is defined as follows.

**Definition 1.** *For an encoding matrix $\boldsymbol{E} \in \mathbb{R}^{K\times N}$, the normalized worst-case squared error (the error) with $S$ stragglers is defined as*

$$\text{err}(\boldsymbol{E}, S) = \frac{1}{K} \max_{\substack{\mathcal{F}\subset[N] \\ |\mathcal{F}|=N-S}} \|\boldsymbol{E}_{\mathcal{F}}\boldsymbol{v}_{\text{opt}} - \boldsymbol{1}_K\|_2^2. \tag{1}$$

When the encoding matrix $\boldsymbol{E}$ is random, the above error is a random variable. We will study the high probability upper bound on the random error.

## III. Existing Constructions

This section summarizes the constructions of four existing gradient codes.

### A. Fractional Repetition Code

The *Fractional Repetition Code* (FRC) is the first gradient code proposed in [13].

The encoding matrix of an $(N, K, L, R)$-FRC gradient code is given by

$$\boldsymbol{E}^{\mathrm{FRC}} = \begin{bmatrix} \boldsymbol{J}_{L\times R} & \boldsymbol{0}_{L\times R} & \cdots & \boldsymbol{0}_{L\times R} \\ \boldsymbol{0}_{L\times R} & \boldsymbol{J}_{L\times R} & \cdots & \boldsymbol{0}_{L\times R} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{0}_{L\times R} & \boldsymbol{0}_{L\times R} & \cdots & \boldsymbol{J}_{L\times R} \end{bmatrix}.$$

The empirical analysis demonstrates that the average error of the FRC is commendable. Nevertheless, it is vulnerable to adversarial stragglers, which gives the exact error:

$$\mathrm{err}(\boldsymbol{E}^{\mathrm{FRC}}, S) = \frac{L}{K}\left\lfloor \frac{S}{R} \right\rfloor.$$

### B. Bernoulli Gradient Code and its regularized variant

To address the drawback inherent in the FRC, a probabilistic gradient code called the *Bernoulli Gradient Code* (BGC) is proposed in [14]. Each entry in the encoding matrix $\boldsymbol{E}^{\mathrm{BGC}}$ of an $(N, K, L)$-BGC is generated i.i.d. according to a Bernoulli distribution $\boldsymbol{E}^{\mathrm{BGC}}_{ij} \sim \mathrm{Bern}(\frac{L}{K})$.

To restrict the computation load per worker to a maximum of $2L$, *regularized Bernoulli Gradient Code* (rBGC) is proposed as a variant of BGC. An $(N, K, L)$-rBGC begins with the same initial structure as an $(N, K, L)$-BGC and then sets entries to zero in columns that exceed $2L$ non-zero entries until only $L$ non-zero entries remain.

### C. BIBD Gradient Code

A $(v, b, k, r, \lambda)$-BIBD is characterized by a set $\mathcal{V}$ of $v$ points and a collection $\mathcal{B}$ of $b$ blocks, each containing exactly $k$ points, such that each point appears in exactly $r$ blocks and every pair of distinct points is included in exactly $\lambda$ blocks [15]. The incidence matrix $\boldsymbol{M}$ for this design is a $v \times b$ binary matrix, where $\boldsymbol{M}_{ij} = 1$ if point $i$ is in block $j$, and $\boldsymbol{M}_{ij} = 0$ otherwise. A BIBD is *symmetric* when $\boldsymbol{M} = \boldsymbol{M}^T$. Given a BIBD with incidence matrix $\boldsymbol{M}$, its dual-design is defined by a design with incidence matrix $\boldsymbol{M}^T$. Throughout the paper, we only consider symmetric BIBD design or dual BIBD design.

Given the incidence matrix $\boldsymbol{M}$ of a BIBD design, a gradient code $(N = b, K = v, L = k, R = r)$-GC is constructed by setting the encoding matrix $\boldsymbol{E}^{\mathrm{B}} = \boldsymbol{M}$. A sufficient condition for the existence of such code are: for $N, K, L, R, \lambda \in \mathbb{N}^+$, $NL = KR$ and $R(L-1) = \lambda(K-1)$. This code is known to be robust against adversarial stragglers. The corresponding worst-case error when there are $S$ stragglers is given by

$$\mathrm{err}(\boldsymbol{E}^{\mathrm{B}}, S) = 1 - \frac{1}{K}\frac{L^2(N-S)}{L + \lambda(N-S-1)}. \tag{2}$$

Moreover, it is known that for any straggling pattern $\mathcal{F}$ with $|\mathcal{F}| = N - S$, the corresponding optimal decoding vector $\boldsymbol{v}_{\mathrm{opt}}(\boldsymbol{E}^{\mathrm{B}}_{\mathcal{F}})$ is the same constant vector and the corresponding squared error is identical.

### D. Soft BIBD Gradient Code

A Soft BIBD gradient code arises from the aim to extend the range of system parameters inherent in BIBD gradient codes, while preserving its superior error performance [9]. An $(N, K, L, R)$-Soft BIBD employs a probabilistic approach to generate its encoding matrix, denoted by $\boldsymbol{E}^{\mathrm{soft}}$, which approximates the encoding matrix of an $(N, K, L, R, \lambda)$-BIBD gradient code.

It is shown in [9] that the average error of the soft BIBD gradient code can be as good as that of BIBD with the same parameters. However, the construction of soft BIBD requires the encoding matrix to have binary entries. This potentially limits the set of system parameters the gradient code can take because the encoding matrix can take any real-valued entries in the general gradient code framework. So, in the remaining part of this paper, we demonstrate how the set of system parameters can be enlarged by exploiting real-valued encoding matrices.

## IV. Sparse Gaussian Gradient Code

### A. Construction

The encoding matrix $\boldsymbol{E}^{\mathrm{SG}}$ is constructed by the element-wise product of two $K \times N$ matrices $\boldsymbol{X}$ and $\boldsymbol{B}$, i.e.,

$$\boldsymbol{E}^{\mathrm{SG}} = \boldsymbol{X} \odot \boldsymbol{B}, \tag{3}$$

where $\odot$ denotes the element-wise multiplication, i.e., $(\boldsymbol{X} \odot \boldsymbol{B})_{ij} = \boldsymbol{X}_{ij}\boldsymbol{B}_{ij}$.

To generate $\boldsymbol{X}$, we choose a correlated multivariate normal distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ as the joint distribution for each row, where $\boldsymbol{\mu} \in \mathbb{R}^N$ and $\boldsymbol{\Sigma} \in \mathbb{R}^{N\times N}$ defined as

$$\boldsymbol{\mu} = a\boldsymbol{1}_N \tag{4}$$

and

$$\boldsymbol{\Sigma} = \begin{bmatrix} b & c & \cdots & c \\ c & b & \cdots & c \\ \vdots & \vdots & \ddots & \vdots \\ c & c & \cdots & b \end{bmatrix}. \tag{5}$$

The $K$ row vectors

$$[\boldsymbol{X}_{i1}, \boldsymbol{X}_{i2}, \ldots, \boldsymbol{X}_{iN}], \quad i \in [K],$$

are generated i.i.d. according $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Note that different rows of $\boldsymbol{X}$ are independent, while different columns of $\boldsymbol{X}$ are correlated due to the correlation in $\boldsymbol{\Sigma}$.

Now, to generate matrix $\boldsymbol{B}$, each entry $\boldsymbol{B}_{ij} \overset{\text{i.i.d.}}{\sim} \mathrm{Bern}\,(\gamma)$. This way, we ensure that a $\gamma$ fraction of the encoding matrix is non-zero in expectation.

For the construction of an $(N, K, L, R, \lambda, \gamma)$-SG gradient code which simulates an $(N, K, L, R, \lambda)$-BIBD gradient code, three criteria have to be satisfied.

1) To mimic the BIBD property that each column has $L$ ones, we require for any $i \in [K]$ and $j \in [N]$,

$$\mathbb{E}\left[\boldsymbol{E}_{ij}^{\mathrm{SG}}\right] = L/K; \tag{6}$$

2) To mimic the BIBD property that the intersection between any two distinct columns is $\lambda$, we require for any $i \in [K]$ and $j, k \in [N]$, $j \neq k$,

$$\mathbb{E}\left[\boldsymbol{E}_{ij}^{\mathrm{SG}}\boldsymbol{E}_{ik}^{\mathrm{SG}}\right] = \lambda/K; \tag{7}$$

3) To ensure the intersections between any column and itself is $L$, we require for any $i \in [K]$ and $j \in [N]$,

$$\mathbb{E}\left[\boldsymbol{E}_{ij}^{\mathrm{SG}}\boldsymbol{E}_{ij}^{\mathrm{SG}}\right] = L/K. \tag{8}$$

Subsequently, for $i \in [K]$ and $j, k \in [N]$, $j \neq k$: $\mathbb{E}\left[\boldsymbol{X}_{ij}\boldsymbol{B}_{ij}\right] = L/K$, $\mathbb{E}\left[\boldsymbol{X}_{ij}\boldsymbol{X}_{ik}\boldsymbol{B}_{ij}\boldsymbol{B}_{ik}\right] = \lambda/K$ and $\mathbb{E}\left[\boldsymbol{X}_{ij}^2\boldsymbol{B}_{ij}^2\right] = L/K$ can be obtained from Equations (6), (7) and (8), respectively. Since the entries of the encoding matrix are independent of the i.i.d. Bernoulli random variables, a system of equations is shown as follows:

$$\begin{cases} a\gamma \overset{(a)}{=} L/K \\ (c+a^2)\gamma^2 \overset{(b)}{=} \lambda/K \\ (b+a^2)\gamma \overset{(c)}{=} L/K \end{cases}, \tag{9}$$

where (a), (b) and (c) follow from $\mathbb{E}\left[\boldsymbol{X}_{ij}\right] = a$, $\mathbb{E}\left[\boldsymbol{X}_{ij}\boldsymbol{X}_{ik}\right] = \mathrm{Cov}(\boldsymbol{X}_{ij}\boldsymbol{X}_{ik}) + \mathbb{E}\left[\boldsymbol{X}_{ij}\right]\mathbb{E}\left[\boldsymbol{X}_{ik}\right] = c + a^2$ and $\mathbb{E}\left[\boldsymbol{X}_{ij}^2\right] = \mathrm{Var}(\boldsymbol{X}_{ij}) + \mathbb{E}\left[\boldsymbol{X}_{ij}\right]^2 = b + a^2$, respectively. Hence, we have

$$\begin{cases} a = L/(K\gamma) \\ b = (\frac{L\gamma}{K} - \frac{L^2}{K^2})/\gamma^2 \\ c = (\frac{\lambda}{K} - \frac{L^2}{K^2})/\gamma^2 \end{cases}, \tag{10}$$

where $\gamma$ is restricted to some range, which will be discussed in the following subsection.

### *B. Existence of the Sparse Gaussian Gradient Code*

In the following, we provide a sufficient condition for the existence of the above Sparse Gaussian gradient code.

**Theorem 1.** *An $(N, K, L, R, \lambda, \gamma)$ Sparse Gaussian gradient code exists if $L \leq K$ and*

$$\max\left\{\frac{\lambda}{L}, \frac{(N-1)(L^2 - K\lambda) + L^2}{KL}\right\} \leq \gamma \leq 1.$$

Guided by Theorem 1, Fig. 1 provides a visualization of a feasible parameter space for $L$ and $R$ in sparse Gaussian gradient codes with specified values of $N, L/K$ and $R/N$. In comparison, we plot several BIBDs with similar density $L/K$ and the parameter region for soft BIBD gradient codes with the same settings of $N, L/K$ and $R/N$. As illustrated, the parameter region for the proposed Sparse Gaussian gradient code expands the choices of feasible parameters compared to those of the soft BIBD gradient codes and the BIBD gradient codes.

In the following, we establish Theorem 1.

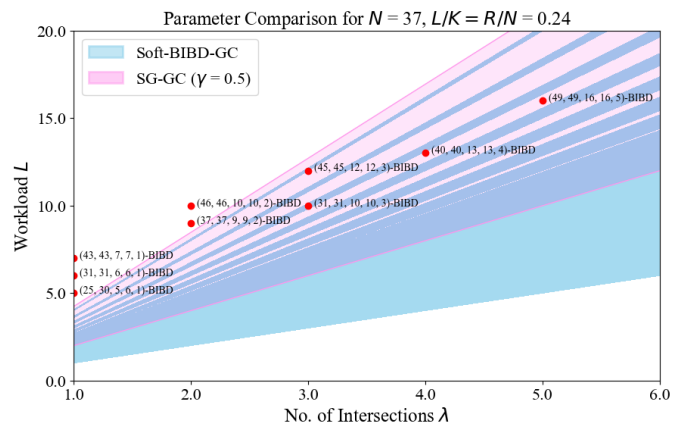


Fig. 1. The region in pink is the parameter region for our proposed SG gradient codes. In contrast, the region in blue is the parameter region for soft-BIBD gradient codes with the probabilistic method. The red dots are combinatorial BIBDs with a similar density, i.e., $\frac{9}{37} \pm 0.05$.

**Fact 1.** *$\lambda$ is an eigenvalue of a square matrix $\boldsymbol{M}$ iff $c + \lambda$ is an eigenvalue of $c\boldsymbol{I} + \boldsymbol{M}$.*

**Lemma 1.** *Any symmetric square matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{N\times N}$ of the form*

$$\boldsymbol{\Sigma} = (b-c)\boldsymbol{I}_N + c\boldsymbol{J}_{N\times N}, \text{ where } b, c \in \mathbb{R},$$

*has eigenvalues $b - c$ with algebraic multiplicity $N - 1$ and $b + (N-1)c$ with algebraic multiplicity 1.*

**Fact 2.** *A multivariate normal distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\mu} \in \mathbb{R}^N$ and $\boldsymbol{\Sigma} \in \mathbb{R}^{N\times N}$, exists if the covariance matrix $\boldsymbol{\Sigma}$ is symmetric and positive semi-definite.*

**Fact 3.** *A matrix $\boldsymbol{\Sigma}$ is positive semi-definite iff all its eigenvalues are nonnegative.*

*Proof of Theorem 1.* (Sketch) Condition $L \leq K$ comes from the parameter requirement of BIBD. For the condition on $\gamma$, note that in the construction of an $(N, K, L, R, \lambda, \gamma)$-SG gradient code, a valid multivariate normal distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ as defined in Equation (5) is needed. According to Lemma 1, the covariance matrix $\boldsymbol{\Sigma}$ has two distinct eigenvalues $b - c$ and $b + (N-1)c$. By Fact 2 and Fact 3, all eigenvalues associated with $\boldsymbol{\Sigma}$ have to be non-negative, i.e. $b - c \geq 0$ and $b + (N-1)c \geq 0$ are required. Plugging in the values of $b$ and $c$ from Equation (10), Theorem 1 follows immediately. □

*Proof of Lemma 1.* Since it is known that the ranks of $\boldsymbol{J}_{N\times N}$ and $c\boldsymbol{J}_{N\times N}$ are 1, they have only one non-zero eigenvalue, and the eigenvalue 0 has algebraic multiplicity $N - 1$. Let $\boldsymbol{u} = \boldsymbol{1}_N$, we have $c\boldsymbol{J}_{N\times N}\boldsymbol{u} = cN\boldsymbol{u}$. Thus, the non-zero eigenvalue is $cN$. For any $(b-c)\boldsymbol{I}_N$, the eigenvalues are $b - c$ with algebraic multiplicity $N$. Therefore, based on Fact 1, $\boldsymbol{\Sigma} \in \mathbb{R}^{N\times N}$ has eigenvalues $b - c$ with algebraic multiplicity $N - 1$ and $b + (N-1)c$ with algebraic multiplicity 1. □

## V. Performance Evaluation

In this section, we provide a high probability upper bound on the *random* worst-case error of the sparse Gaussian gradient

code. In the regime of parameters specified in Theorem 2, the proposed sparse Gaussian gradient code can have an error performance comparable to that of the symmetric or dual BIBD gradient code with the same parameters.

**Theorem 2.** *Consider $(N, K, L, R, \lambda, \gamma)$ Sparse Gaussian gradient codes with encoding matrix $\boldsymbol{E}^{\mathrm{SG}}$ and $K = \Theta(N), L = \Theta(N), \lambda = \Theta(N), \gamma = \Theta(1)$. For $0 < \delta < 1/2$ and $0 < \alpha < 2\delta$, let the number of stragglers $S = O(N^{\alpha})$ and $\rho = \frac{L}{L+\lambda(N-S-1)}$. If $\rho^2(N - S)(b + (N - S - 1)c) < 2$ and $c \geq 0$, then*

$$\mathbb{P}\left[\mathrm{err}(\boldsymbol{E}^{\mathrm{SG}}, S) \leq \mathrm{err}(\boldsymbol{E}^{\mathrm{B}}, S) + \frac{1}{K^{1/2-\delta}}\right] \geq 1 - e^{-\Theta(N^{2\delta})},$$

*where $\mathrm{err}(\boldsymbol{E}^{\mathrm{B}}, S)$ is the worst-case error of symmetric or dual $(N, K, L, R, \lambda)$-BIBD gradient code given in Equation (2).*

**Remark 1.** *Note that $\frac{1}{K^{1/2-\delta}} = o(1)$ and $e^{-\Theta(N^{2\delta})} = o(1)$. This means in the specified parameter regime, with high probability, the error associated with the sparse Gaussian gradient code concentrates at or below the error of the BIBD Gradient Code with the same parameters.*

We sketch the proof of Theorem 2 as follows. We first upper bound the worst-case error by the error corresponding to the constant decoding vector $\rho\mathbf{1}$. For any non-straggler set $\mathcal{F} \subset [N]$ with $|\mathcal{F}| = N - S$, we have

$$\begin{aligned}\mathrm{err}(\boldsymbol{E}^{\mathrm{SG}}, S) &\leq \tfrac{1}{K} \max_{\mathcal{F}} \|\rho \boldsymbol{E}_{\mathcal{F}} \mathbf{1}_{N-S} - \mathbf{1}_K\|_2^2 \\ &= \tfrac{1}{K} \max_{\mathcal{F}} \textstyle\sum_{i=1}^{K} X_i^2,\end{aligned}$$

where $X_i = \rho \sum_{j \in \mathcal{F}} \boldsymbol{E}_{ij}^{\mathrm{SG}} - 1$ for $i \in [K]$.

Then, we show that $X_i^2$'s are *sub-exponential* random variables defined as follows.

**Definition 2** ( [16, Definition 2.7.5])**.** *The sub-exponential norm of a random variable $Z$ is defined as*

$$\|Z\|_{\psi_1} = \inf\{t > 0 : \mathbb{E}\left[\exp(|Z|/t) \leq 2\right]\}.$$

*If $\|Z\|_{\psi_1}$ is finite, we say that $Z$ is sub-exponential.*

**Lemma 2.** *Let $\sigma_{\max} = \rho(N - S)^{1/2}(b + (N - S - 1)c)^{1/2}$. If $\sigma_{\max} < \sqrt{2}, c \geq 0$ and $\rho \geq 0$, then for any $i \in [K]$, $X_i^2$ is a sub-exponential random variable with sub-exponential norm*

$$\|X_i^2\|_{\psi_1} \leq \max\left\{2\sigma_{\max}^2 + \frac{\beta_{\max}^2}{\ln\left(\sqrt{2}/\sigma_{\max}\right)}, 4\sigma_{\max}^2\right\},$$

*where $\beta_{\max} = \rho(N - S)a - 1$.*

Next, we subtract the mean and let $Z_i \triangleq X_i^2 - \mathbb{E}\left[X_i^2\right]$. From [16, p.36], it can be shown that $\|X_i^2 - \mathbb{E}\left[X_i^2\right]\|_{\psi_1} \leq C\|X_i^2\|_{\psi_1}$ for an absolute constant $C > 0$, which implies that $Z_i$ is a sub-exponential random variable with mean zero. We also observe that $Z_1, \ldots, Z_K$ are i.i.d. because each row of the encoding matrix is independently sampled from the same designated multivariate Gaussian distribution (cf. Section IV-A). Moreover, by choosing an appropriate constant $\rho$, it can be shown that $\mathbb{E}\left[\frac{1}{K}\sum_{i=1}^{K} X_i^2\right]$ equals the error of BIBD with the same parameters.

**Lemma 3.** *Let $\rho = \frac{L}{L+\lambda(N-S-1)}$. For any $\mathcal{F}$ with $|\mathcal{F}| = N - S$, the expected error using constant decoding vector $\rho\mathbf{1}_{|\mathcal{F}|}$ is given by*

$$\mathbb{E}\left[\left\|\rho \boldsymbol{E}_{\mathcal{F}} \mathbf{1}_{|\mathcal{F}|} - \mathbf{1}_K\right\|_2^2\right] = \mathbb{E}\left[\textstyle\sum_{i=1}^{K} X_i^2\right] = K \cdot \mathrm{err}(\boldsymbol{E}^{\mathrm{B}}, S),$$

*where $\mathrm{err}(\boldsymbol{E}^{\mathrm{B}}, S)$ is the error of an $(N, K, L, R, \lambda)$-BIBD gradient code.*

Finally, for each $\mathcal{F}$, we employ Bernstein's inequality to establish the concentration of the (random) error using the constant decoding vector. Taking union bound over all $\mathcal{F}$ with $|\mathcal{F}| = N - S$ establishes Thoerem 2.

**Theorem 3** (Bernstein's inequality [16, Theorem 2.8.1])**.** *Let $Z_1, \ldots, Z_K$ be independent, mean zero, sub-exponential random variables. Then, for every $t > 0$, we have*

$$\begin{aligned}&\mathbb{P}\left[\left|\textstyle\sum_{i=1}^{K} Z_i \geq t\right|\right] \\ &\leq 2\exp\left[-C\min\left\{\frac{t^2}{\sum_{i=1}^{K}\|Z_i\|_{\psi_1}^2}, \frac{t}{\max_i \|Z_i\|_{\psi_1}}\right\}\right],\end{aligned}$$

*where $C > 0$ is an absolute constant.*

We now complement our theoretical analysis with a simulation comparing the worst-case error among five gradient codes with similar workload densities. As Fig. 2 shown, the proposed SG gradient code performs comparably to the BIBD gradient code and outperforms FRC, BGC, and rBGC gradient codes. This means the proposed SG gradient code can be a better alternative than the other three when the BIBD gradient code of the same parameter does not exist.

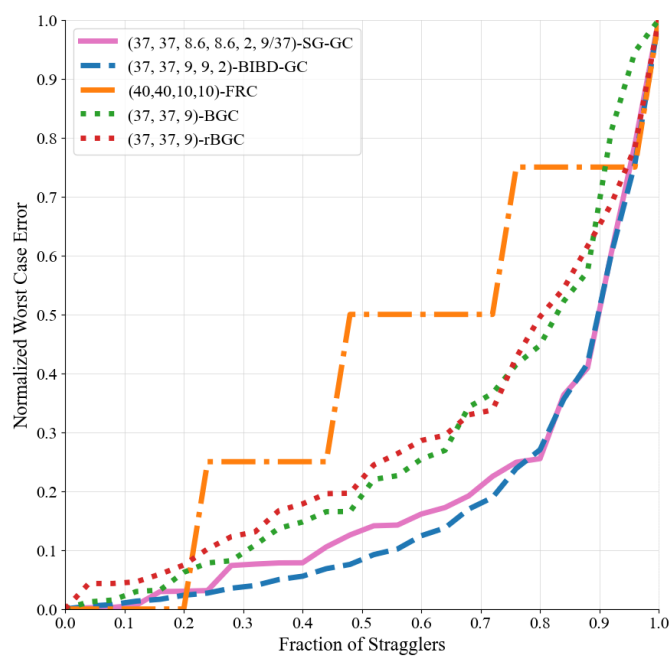


Fig. 2. A comparison of worst-case normalized errors between our proposed SG gradient code and some existing gradient codes which were discussed in the paper, with a specific focus on similar column densities, being 0.25 for the FRC and $\frac{9}{37}$ for the remaining gradient codes.

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